

Course Overview

This course is designed for Cloud Engineers, System Administrators, DevOps Engineers, Build & Release Engineers, Software Developers, Azure Administrators, and IT Professionals who want to implement modern DevOps practices using Azure DevOps and GitHub.

The course follows a practical approach with hands-on labs, real-world deployment scenarios, Infrastructure as Code (IaC), DevSecOps, Monitoring, CI/CD implementation, and AZ-400 certification preparation.

Month 1: Foundations & Planning

Module 1: Cloud, Azure & DevOps Fundamentals

Cloud Fundamentals

- What is Cloud Computing
- Cloud Service Models (IaaS, PaaS, SaaS)
- Public, Private & Hybrid Cloud
- Benefits of Cloud Computing

Azure Fundamentals

- Azure Portal Overview
- Resource Groups
- Virtual Machines
- Storage Accounts
- Networking Basics
- Azure Subscription Management

DevOps Fundamentals

- What is DevOps
 - DevOps Lifecycle
 - SDLC Overview
 - Agile Methodology
 - Scrum Framework
 - CI/CD Concepts
 - DevOps Toolchain
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Module 2: Linux & Git Fundamentals

Linux Fundamentals

- Linux Architecture
- Linux File System
- Linux Commands
- Users & Groups
- Permissions
- Package Management
- Services Management

Git Fundamentals

- Version Control Systems
- Distributed Version Control
- Git Installation
- GitHub Account Setup
- Clone
- Commit
- Push
- Pull

Module 3: Azure DevOps Fundamentals

Azure DevOps Overview

- Azure DevOps Services
- Azure DevOps Organization
- Azure DevOps Projects
- Teams & Permissions
- Access Levels
- Microsoft Entra ID Integration
- Service Connections

Azure DevOps Services

- Azure Boards
 - Azure Repos
 - Azure Pipelines
 - Azure Artifacts
 - Azure Test Plans
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Module 4: Azure Boards

Work Management

- Epics
- Features
- User Stories
- Tasks
- Backlogs

Agile Planning

- Scrum Process
- Agile Process
- Sprint Planning
- Capacity Planning
- Kanban Boards

Reporting

- Queries
- Dashboards
- Analytics Views
- Work Item Traceability

Month 2: Source Control & Continuous Integration

Module 5: Git & Branching Strategies

Git Operations

- Branches
- Merge
- Rebase
- Conflict Resolution
- Tags
- Revert
- Reset
- Cherry Pick

Branching Strategies

- GitHub Flow
 - Feature Branch Workflow
 - Trunk-Based Development
 - Release Branches
 - Hotfix Branches
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Module 6: Azure Repos & GitHub

Azure Repos

- Repository Management
- Import Repositories
- Git Credential Manager

Collaboration

- Pull Requests
- Code Reviews
- Build Validation
- Branch Policies
- Protected Branches

GitHub Fundamentals

- GitHub Repositories
 - GitHub Projects
 - GitHub Issues
 - GitHub Discussions
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Module 7: Azure Pipelines (CI)

Continuous Integration

- CI Concepts
- Pipeline Architecture
- Pipeline Triggers

YAML Pipelines

- Pipeline Structure
- Stages
- Jobs
- Steps
- Variables

- Conditions

Advanced Pipelines

- Templates
- Parameters
- Multi-Stage Pipelines
- Deployment Jobs
- Environments
- Approvals

Agents

- Microsoft Hosted Agents
 - Self Hosted Agents
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Module 8: GitHub Actions

GitHub Actions Fundamentals

- Workflow Files
- Events & Triggers
- Jobs
- Steps

Runners

- GitHub Hosted Runners
- Self Hosted Runners

Security

- Secrets
 - Variables
 - OIDC Authentication
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Month 3: Security, CD, IaC & Monitoring

Module 9: Azure Artifacts

Package Management

- Azure Artifacts Overview

- Feeds
 - Upstream Sources
 - Package Publishing
 - Package Consumption
 - Semantic Versioning
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Module 10: Security & Azure Key Vault

Security Fundamentals

- Authentication vs Authorization
- Service Principals
- Managed Identities
- PAT Tokens
- Service Connections

Azure Key Vault

- Secrets
- Keys
- Certificates
- Variable Groups
- Pipeline Integration

DevSecOps

- Secret Scanning
 - Dependency Scanning
 - CodeQL
 - Dependabot
 - Defender for DevOps
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Module 11: Continuous Delivery

Release Management

- Continuous Delivery Concepts
- Deployment Gates
- Environment Approvals
- Release Governance

Application Deployment

- Azure App Service

- Azure Virtual Machines
 - Azure Container Apps
 - Azure Kubernetes Service (AKS)
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Module 12: Deployment Strategies

Modern Deployment Strategies

- Blue-Green Deployment
 - Canary Deployment
 - Rolling Deployment
 - Feature Flags
 - Rollback Strategy
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Module 13: Infrastructure as Code (IaC)

Azure CLI

- Resource Deployment
- Automation

Bicep

- Fundamentals
- Parameters
- Variables
- Modules

Terraform

- Terraform Architecture
- State Management
- Remote Backend
- Modules

DevOps Integration

- Terraform Pipelines
 - Bicep Pipelines
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Module 14: Testing & Quality Management

Azure Test Plans

- Test Plans
- Test Suites
- Test Cases

Testing Concepts

- Unit Testing
- Integration Testing
- Functional Testing
- Regression Testing

Quality Gates

- Build Validation
- Code Coverage
- Release Gates

Module 15: Monitoring & Observability

Azure Monitor

- Metrics
- Alerts
- Action Groups

Log Analytics

- Log Collection
- Log Analysis

Application Insights

- Distributed Tracing
- Dependency Tracking
- Live Metrics

KQL Basics

- Log Queries
- Troubleshooting